

AYMANE AIT DADS

ML/NLP Research Intern | LLM Fine-Tuning · Reasoning · Conversational AI

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT on five structured reasoning categories. Achieved 1st place in the EURECOM class leaderboard for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 fine-tuning with ensemble methods, and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark. Brings hands-on expertise in LLM training and finetuning techniques, retrieval augmented generation, chain-of-thought reasoning, agentic pipelines, and representation learning directly aligned with Samsung AIC's adaptive personalized AI research agenda. Aims to contribute novel deep learning algorithms for natural language processing and reasoning on-device, advancing Samsung's vision of fast, accurate, and contextual AI interactions.

CORE COMPETENCIES

LLM Training & Finetuning | Retrieval Augmented Generation | Reasoning & Action Planning | Natural Language Processing | Agent AI & Tool Use | Representation Learning | Multimodal AI | On-Device AI

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a **Random Forest + K-Means clustering pipeline** on 100K+ fiber-optic network records to classify quality across thousands of nodes, directly supporting the network optimization team.
- Designed a customer performance profiling system mapping user metrics to **5 commercial service tiers**, operationalizing model outputs into actionable network maintenance recommendations.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and end-to-end pipeline implementation using Python and Scikit-learn.
- Delivered stakeholder presentations translating **complex model outputs** into business-ready maintenance recommendations, demonstrating communication skills across technical and non-technical audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT ($r=32$, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation), building a diagnostic evaluation pipeline that revealed reasoning over-expansion - not incorrect reasoning - as the primary failure mode driving hard-task errors.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and 10-view TTA across 250K images from 25+ generator types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** for out-of-distribution generalization, and built a 3-model ensemble weighted by validation AUC using FP16 mixed precision training.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

Twitter Sentiment Analysis

EURECOM NLP Course · 2025

- Built an **end-to-end natural language processing pipeline** covering tokenization, TF-IDF, word2vec embeddings, and transformer fine-tuning for multi-class sentiment classification.
- Applied **representation learning** techniques including semantic embeddings and hyperparameter optimization, directly mapping to conversational AI and language model research objectives.

Python · Hugging Face Transformers · Scikit-learn · word2vec

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM	<i>Sep 2023 - Present</i>
Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Image Security, Cloud Computing, Statistics	
CPGE - Mathematics & Physics - Ibn Timiya	<i>2021 - 2023</i>

SKILLS

LLM Fine-Tuning & Reasoning: LoRA / QLoRA | SFTTrainer | chain-of-thought | vLLM | Mamba-Transformer | RAG | agentic pipelines
ML / NLP / Deep Learning: PyTorch | Hugging Face Transformers | PEFT | TRL | Scikit-learn | CLIP / ViT | sentiment analysis
MLOps & Infrastructure: Python | BF16 / FP16 mixed precision | Triton | Git | Linux | Docker | Kaggle GPU